

SAMARKAND, Uzbekistan

WMO: 386960

Lat: 39.701N

Lon: 66.984E

Elev: 678

StdP: 93.44

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.1	-14.1	1.2	-7.1	-11.2	1.6	-2.0	9.6	3.6	8.4	3.8	2.1	140	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	36.9	19.0	35.4	18.5	34.2	18.1	20.7	33.1	19.9	32.3	19.2	31.6	3.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.9	25.3	15.5	11.9	24.4	14.6	11.3	24.0	63.1	32.9	60.0	32.5	57.5	31.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.0	-12.3	39.7	4.2	0.9	-15.3	40.3	-17.8	40.9	-20.1	41.4	-23.2	42.1		
(5)			-12.8	22.5	4.2	1.2	-15.8	23.4	-18.2	24.1	-20.6	24.8	-23.6	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	2.7	4.0	9.8	15.1	20.7	25.1	26.7	25.6	21.0	14.9	8.6	4.0
	DBStd	9.54	5.12	5.59	5.17	4.68	3.45	2.65	2.36	2.51	3.23	4.33	4.99	4.91
	HDD10.0	771	231	176	66	12	0	0	0	0	0	11	82	192
	HDD18.3	2156	486	401	267	115	17	1	0	1	13	120	294	443
	CDD10.0	2562	4	9	60	165	331	452	518	483	329	163	40	8
	CDD18.3	906	0	0	3	19	89	203	259	225	92	13	2	0
	CDH23.3	11071	0	1	33	205	927	2511	3401	2755	1021	196	21	0
	CDH26.7	5507	0	0	6	53	333	1288	1903	1476	398	48	3	0
Wind	WSAvg	2.8	2.4	2.8	3.0	3.1	3.1	3.0	2.9	2.8	2.9	2.5	2.3	2.4
Precipitation	PrecAvg	392	46	54	74	62	36	7	2	2	3	16	43	48
	PrecMax	579	86	119	134	143	106	20	18	18	13	43	124	114
	PrecMin	181	15	16	26	4	1	0	0	0	0	0	2	1
	PrecStd	97	18	28	29	31	30	7	4	4	3	12	30	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	21.0	27.0	30.9	34.1	37.7	39.2	38.0	34.6	31.0	26.1	19.4
		MCWB	8.2	10.3	12.8	16.2	18.1	19.0	19.6	19.3	17.8	15.8	12.9	8.7
	2%	DB	14.6	17.0	23.2	28.1	32.0	35.8	37.5	36.1	32.8	27.8	21.9	16.1
		MCWB	7.4	8.9	12.2	15.9	17.3	18.4	19.2	18.8	17.2	14.3	11.3	8.2
	5%	DB	12.1	14.2	20.8	25.8	30.1	34.2	35.9	34.9	30.9	25.3	18.9	13.8
		MCWB	6.2	7.8	11.6	15.0	16.9	18.0	18.7	18.4	16.4	13.6	10.5	7.2
	10%	DB	9.9	12.0	18.1	23.2	28.4	32.9	34.2	33.5	29.0	23.2	16.3	11.3
		MCWB	5.4	6.8	10.8	14.0	16.4	17.7	18.3	17.9	15.5	12.8	9.5	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.5	14.8	18.7	20.2	21.6	21.9	21.8	19.9	17.2	14.0	10.3
		MCDB	14.9	18.0	23.0	26.4	29.8	33.2	34.7	33.7	30.6	27.5	21.1	15.8
	2%	WB	8.2	9.9	13.4	17.1	18.8	20.1	20.7	20.3	18.2	15.3	12.6	8.9
		MCDB	13.0	14.9	20.6	24.8	28.4	32.4	34.0	32.7	30.0	25.0	19.7	14.7
	5%	WB	6.9	8.7	12.3	16.0	17.9	19.2	19.9	19.5	17.0	14.3	11.3	7.7
		MCDB	11.0	13.2	18.8	23.6	27.7	31.4	33.1	32.1	28.9	23.5	17.7	12.7
	10%	WB	5.7	7.5	11.2	14.9	17.1	18.5	19.1	18.6	16.1	13.4	10.1	6.5
		MCDB	9.2	11.4	17.2	21.2	26.4	30.4	32.2	31.4	27.5	21.7	15.4	10.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.5	11.2	12.7	14.2	14.7	15.1	14.6	13.4	10.7	9.2
		MCDBR	12.0	12.9	14.2	14.7	14.9	16.2	16.4	16.8	16.8	16.4	14.8	13.0
		MCWBR	6.9	7.2	6.8	7.1	6.4	6.5	6.1	6.4	7.2	7.5	7.4	7.4
	5% WB	MCDBR	11.0	11.3	12.6	12.6	13.5	14.7	15.5	16.0	15.8	14.6	12.9	11.7
MCWBR		6.7	6.9	6.8	6.8	6.5	6.5	6.4	6.6	7.4	7.3	6.9	7.1	
Clear-Sky Solar Irradiance	taub	0.355	0.398	0.464	0.482	0.475	0.467	0.448	0.427	0.412	0.450	0.412	0.357	
	taud	2.187	2.073	1.938	1.930	1.989	2.012	2.064	2.111	2.099	1.949	2.035	2.160	
	Ebn at Noon	782	796	780	797	813	818	829	837	822	723	696	741	
	Edh at Noon	104	134	171	184	177	173	163	151	143	150	120	99	
All-Sky Solar Radiation	RadAvg	1.85	2.61	3.74	5.21	6.65	7.66	7.62	6.90	5.56	3.73	2.25	1.67	
	RadStd	0.23	0.35	0.35	0.55	0.37	0.22	0.24	0.21	0.17	0.27	0.28	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A	+98	+78
(47) Regional (8 neighbors)	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A	+59

Nomenclature: See separate page